

Developmental Paediatrics, Charts and Anthropometry

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Developmental Paediatrics, Charts & Anthropometry

Dr Ijeoma Stella – Paediatrics, topic 3

Outline

Growth vs development · Factors affecting growth/development · Phases of childhood · Patterns of growth (weight, length, head circumference) · Weight/height formulas · Fields of development · Screening vs assessment · Growth assessment & monitoring · Reading the growth chart · Body proportions · Dentition · Developmental milestones · Speech red flags · Draw-a-Person test

Growth vs development

- **Growth** = the **increase in SIZE** of a child
- **Development** = the **progressive acquisition of skills** – physical (motor), cognitive (thought), linguistic (communication) and social (emotional)

Development is an **increase in function and complexity of skills**, whereas growth is an increase in size.

Factors affecting growth and development

- **Biologic** – genetic inheritance, intrauterine infection or teratogens, perinatal accidents, neurologic and hormonal maturation
- **Psychological** – the mother figure, feeding practices, child-rearing environment
- **Social** – family socio-economic status, extended family, cultural practices, geo-political milieu

Phases of childhood

Prenatal · Neonatal · Infant · Preschool · School age · Adolescence

Patterns of growth

Neonatal (first 28 days): usually weighs $\geq 2,500$ g, length > 45 cm. Weight may **drop 3-10% in the first week** (loss of extravascular fluid, low colostrum intake).

Infant (first 12 months):

- **Weight** gain ~ 30 g/day (2nd month), slowing to 20 g/day (3rd-5th). Rule of thumb: an exclusively breastfed term baby **doubles birth weight by ~ 5 months and triples it by 9-12 months**
- **Length** – highest velocity: from ≥ 45 cm at birth to ≥ 66 cm at 1 year (~ 2 cm/month average)
- **Head circumference** – parallels the brain's growth: from **33 cm at birth to 45 cm at 1 year**

Preschool (2-5 yr): velocities ~ 240 g/month, 1 cm/month, 0.25 cm/month (weight, height, HC).

Primary school (6-12 yr): ~ 3 kg/year, 6 cm/year, 0.5 cm/year.

Adolescence (13-18 yr, may span 10-20): three phases – early (10-13), middle (14-16), late (17-20).

Formulas (weight & height up to 12 years)

Age	Weight (kg)	Height (cm)
Birth	3	48
3-12 months	$(\text{age in months} + 9) \div 2$	–
1-6 years	$(\text{age in years} \times 2) + 8$	$(\text{age} \times 6) + 77$
7-12 years	$(\text{age in years} \times 7 - 5) \div 2$	$(\text{age} \times 6) + 77$

Fields of development

Gross motor · Vision and fine motor · Hearing, speech and language (cognitive) · Social, emotional and behavioural (psychological) · Vegetative.

Screening vs assessment

- **Screening** – checks of a **whole population/groups** at set ages by trained professionals
- **Assessment** – **detailed analysis** of overall or specific development

Global developmental delay = delay in **2 or more spheres in a child < 5 years**. After 5 years the term **mental retardation** (delayed IQ) is used.

Growth assessment & monitoring

Parameters: **weight, height, head circumference, mid-arm circumference, skinfold thickness, body proportions, skeletal maturation, dental development.**

Growth monitoring uses the **growth chart** (an international NCHS reference standard) to track **weight-for-age, height-for-age, HC-for-age and weight-for-height** from birth. Physiologic shifts across curves occur with **parental genetics (first 2 years), prematurity** (apply gestational correction until **18 months for HC, 24 months for weight, 40 months for height**), and the **adolescent growth spurt**.

Reading the growth chart (deviation from the norm)

- **Failure to thrive:** weight below the 5th percentile or drops below 2 major percentile lines
- **Acute undernutrition:** weight-for-age and weight-for-height drop, but height-for-age stays normal
- **Stunting:** height-for-age drops, while weight-for-height may return toward normal
- In **nutritional** causes, **weight drops before or with height**; in **constitutional, familial and endocrine** causes, the **height curve is affected first**
- **Obesity:** weight-for-height > **120%** of median, with raised **BMI**

Body proportions

The body is divided at the **symphysis pubis**: upper segment (pubis->crown) and lower (pubis->sole). The **upper:lower (U/L) ratio** is **~1.7 at birth, 1.3 at 3 years, 1.0 after 7 years**.

- **Higher** ratios -> short-limb dwarfism, rickets
- **Lower** ratios -> some adolescents, sickle cell disease

Dentition

Mineralisation begins in the **2nd trimester**; deciduous teeth complete by ~3 years, permanent by ~25 years.

- **Delayed eruption** = no first tooth by **13 months** (familial, idiopathic, **hypothyroidism, rickets**)
- **Early exfoliation** (before the 5th birthday) – idiopathic, trauma, gingivitis, **scurvy, histiocytosis X**

Developmental domains and pattern

Domains: **gross motor, fine motor, social, emotional, communication, cognition.**

When assessing a young child: consider the **age**, focus questions on the area of **likely maximum progress**, offer **suitable toys**, and **observe** how the child uses them and interacts. Instruments: **Denver II**, Bayley scales, PEDS, Paediatric Symptom Checklist, **M-CHAT** (autism screen).

Who assesses and why: parents (reassurance/detection), teachers (school performance), physicians (identify specific delay/disability and plan intervention).

Recognising speech and language problems early

Refer if: **not talking at all by 2 years**; speech **largely unintelligible after 3 years**; dropping beginning consonants after 3 years; **not using 2-3-word sentences by 3 years**; mostly vowel sounds; word endings missing after 5 years; or a **monotone/too-loud/too-soft/poor-quality voice** (may indicate hearing loss).

The Draw-a-Person test

A standardised way to assess **intellectual and emotional state** – first used by **Florence Goodenough (1926)**, validated for **Nigerian children aged 6-11** by Izuora and Ebigbo. Done anywhere with pencil and paper.

- **4 points = 1 “draw-a-person” year**; no points expected before age 4, so **3 points are added**
- **Maturation age = (points $\div$ 4) + 3**
- **Man-drawing quotient = maturation age $\div$ chronologic age**; normal **> 0.75** (i.e. **> 75% = IQ normal**)

References

Lecture material of Dr Ijeoma Stella; standard paediatric growth and development texts.